

1. Lab Overview

In this lab, I configure a **DHCP client** on Windows 10 and configure a **DHCP Server** on Windows Server 2019. The lab focuses on DHCP client settings, DHCP server network configuration, DHCP role installation, DHCP scope creation, DHCP options, scope activation, and DHCP traffic analysis using Wireshark.

This lab helps me understand how DHCP automatically assigns IP addresses and network settings to client computers in a Windows network environment.

2. Business Scenario

This lab simulates a small business or school network where Windows clients need to automatically receive IP configuration from a DHCP server.

As the system administrator, I configure the Windows 10 client to obtain an IP address automatically. Then I configure the Windows Server 2019 system as a DHCP server, create a DHCP scope, activate the scope, renew the client's IP address, and analyze DHCP traffic using Wireshark.

DHCP is important in real IT environments because it reduces manual IP configuration, prevents IP address conflicts, and allows client devices to receive network settings automatically.

3. Lab Objectives

The objectives of this lab are to:

- Configure a Windows 10 system as a DHCP client.
- Configure static IP settings on the Windows Server 2019 DHCP server.
- Install the DHCP Server role on Windows Server 2019.
- Open and use the DHCP console.
- Create a DHCP scope.
- Configure the DHCP IP address range.
- Add exclusions to protect reserved/static IP addresses.
- Configure DHCP lease duration.
- Configure DHCP options such as router/default gateway and DNS server.
- Activate the DHCP scope.
- Release and renew a client IP address.
- Capture and analyze DHCP traffic using Wireshark.
- Verify the DHCP-assigned IP configuration.

Your original lab goals are to configure a DHCP client system, configure a DHCP server system, install the DHCP role, create a DHCP scope, and analyze/test/debug DHCP traffic.

4. Lab Environment

Item	Details
DHCP Client	Windows 10
DHCP Server	Windows Server 2019
Server IP Address	10.0.0.52
Server Subnet Mask	255.0.0.0
Server Preferred DNS Server	127.0.0.1
DHCP Scope Range	10.0.0.1 – 10.0.0.99
Excluded Address	10.0.0.52
Router / Default Gateway Option	10.0.0.200
DNS Server Option	10.0.0.52
Server Name	WARA-SERVER
Tools Used	Server Manager, DHCP Console, Command Prompt, Wireshark

5. Tools Used

- Windows 10
- Windows Server 2019
- Server Manager
- DHCP Console
- Command Prompt
- Wireshark
- IPv4 network adapter settings
- Windows Network and Sharing Center

6. Steps Performed

Step 1: Configure the Windows 10 DHCP Client

I begin by configuring the Windows 10 client to receive network settings automatically.

In the IPv4 properties, I select:

Obtain an IP address automatically

Obtain DNS server address automatically

This allows the Windows 10 system to receive IP address information from the DHCP server instead of using manual static settings.

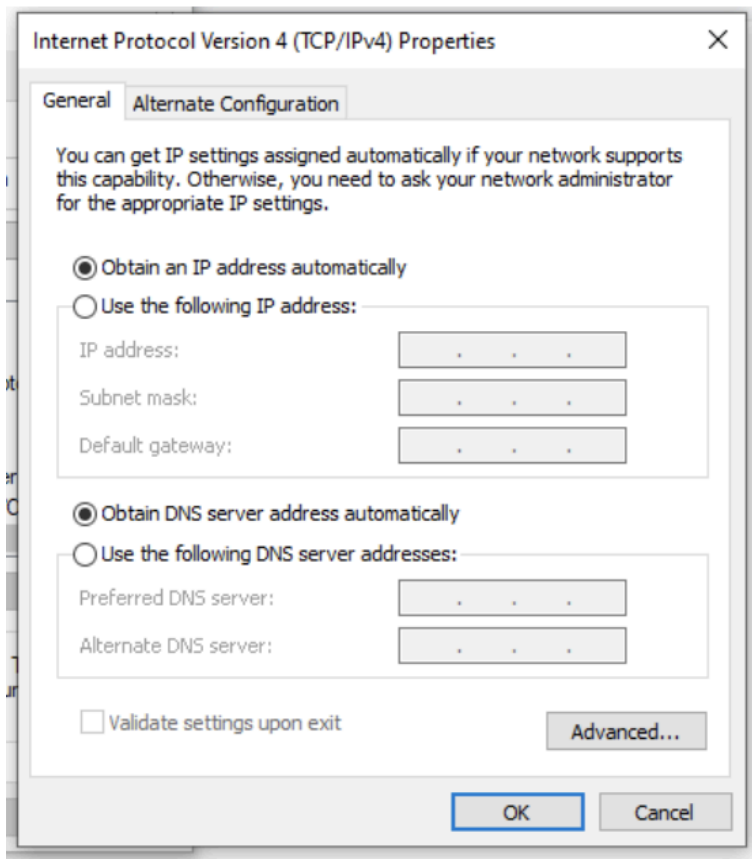
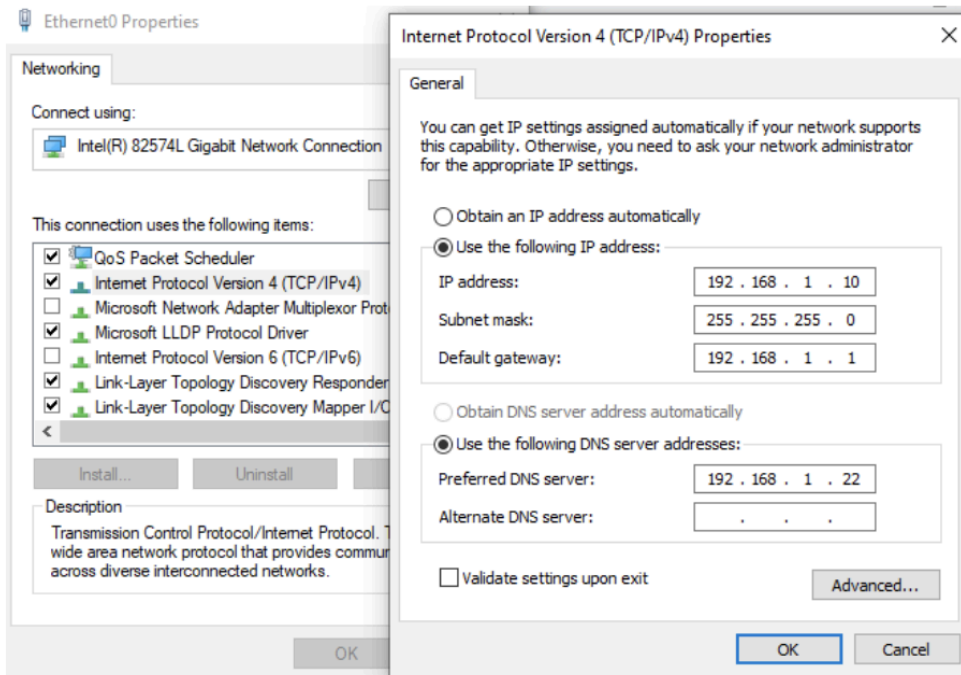


Figure 1: Windows 10 client configured to obtain IP and DNS settings automatically.

Step 2: Configure the Windows Server 2019 DHCP Server Network Settings

Next, I configure the Windows Server 2019 system with a static IP address.

The server uses:

IP address: 10.0.0.52

Subnet mask: 255.0.0.0

Default gateway: Leave blank

Preferred DNS server: 127.0.0.1

A DHCP server should use a static IP address so clients can consistently reach it for IP address assignment.

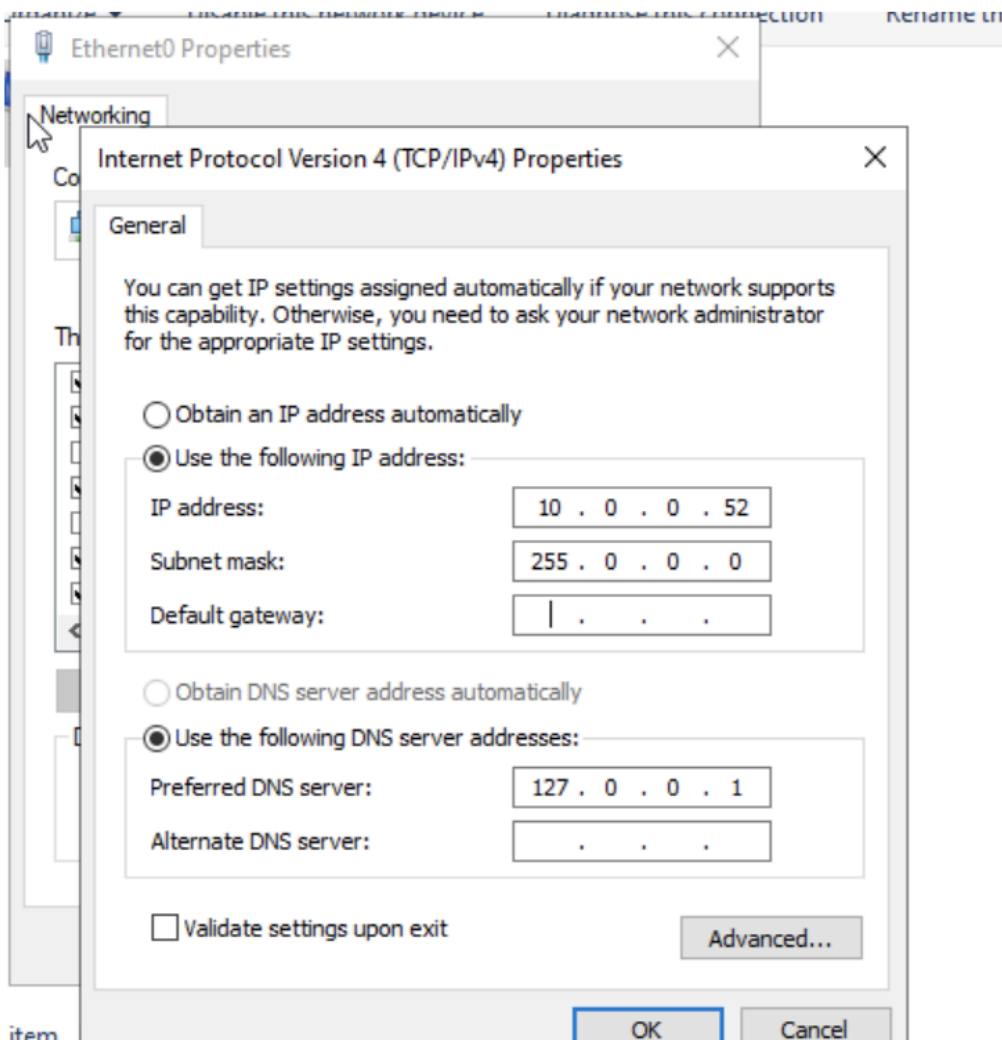


Figure 2: Windows Server 2019 configured with a static IP address for DHCP services.

Step 3: Rename the DHCP Server

I verify or configure the server name as:

WARA-SERVER

Using a clear server name makes the DHCP server easier to identify in Server Manager and the DHCP console.

Device name	WARA-SERVER
Processor	AMD Ryzen 7 5700U with Radeon Graphics 1.80 GHz (2 processors)
Installed RAM	2.00 GB
Device ID	39584965-0E7D-488F-B1A5-3749AA8BC2AE
Product ID	00454-40000-00001-AA737
System type	64-bit operating system, x64-based processor
Pen and touch	No pen or touch input is available for this display

Figure 3: Windows Server 2019 renamed to WARA-SERVER.

Step 4: Install the DHCP Server Role

I open **Server Manager** and use **Add Roles and Features** to install the **DHCP Server** role.

The process includes:

1. Opening Server Manager.
2. Selecting **Add Roles and Features**.
3. Selecting the Windows Server 2019 server.
4. Choosing the **DHCP Server** role.
5. Adding required features.
6. Confirming the role selection.
7. Completing the installation.

Select destination server

DESTINATION SERVER
WARA-SERVER.wara.edu

Before You Begin

Installation Type

Server Selection

Server Roles

Features

Confirmation

Results

Select a server or a virtual hard disk on which to install roles and features.

- ☒ Select a server from the server pool
- ☐ Select a virtual hard disk

Server Pool

Filter:

Name	IP Address	Operating System
WARA-SERVER.wara.edu	10.0.0.52	Microsoft Windows Server 2022 Standard Evaluation

1 Computer(s) found

This page shows servers that are running Windows Server 2012 or a newer release of Windows Server, and that have been added by using the Add Servers command in Server Manager. Offline servers and newly-added servers from which data collection is still incomplete are not shown.

Server Manager

Add Roles and Features Wizard

Dashboard

Select destination server

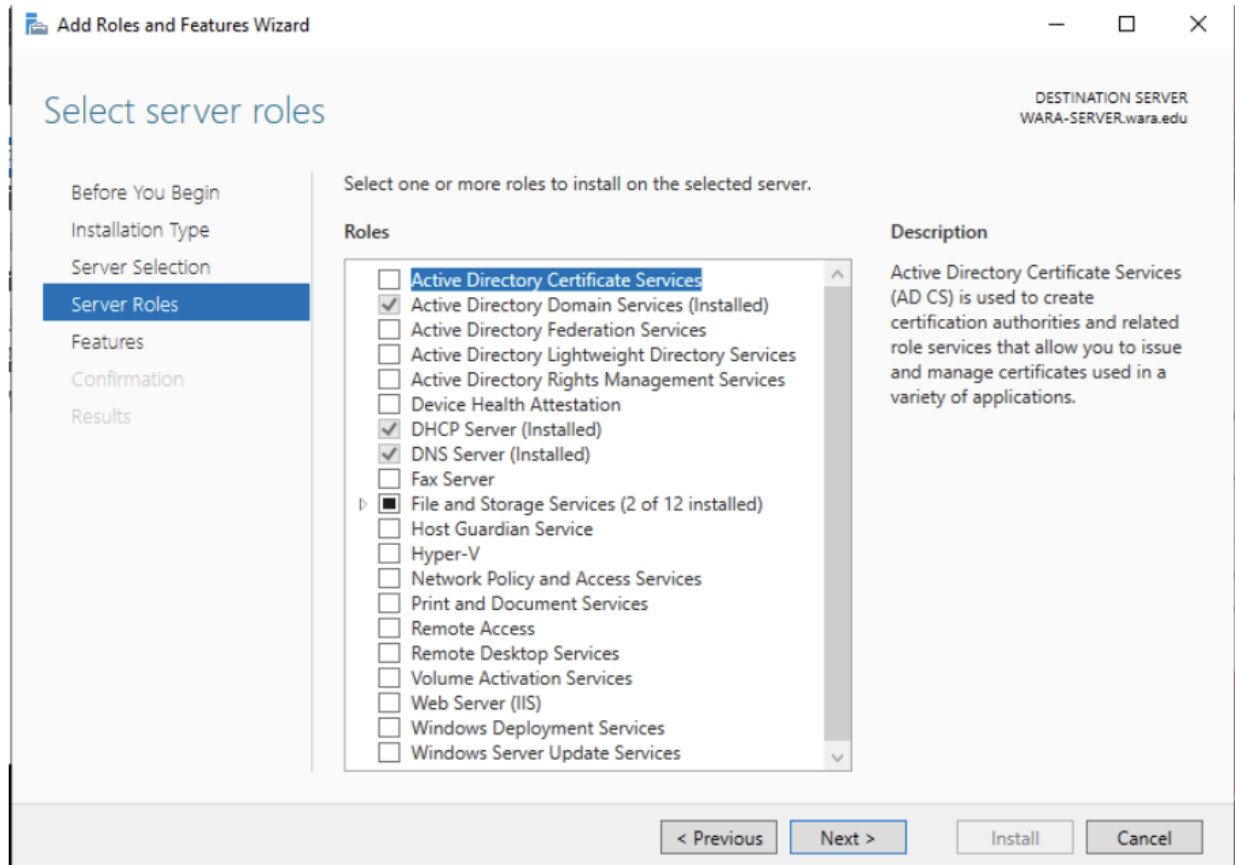


Figure 4: DHCP Server role selected for installation on Windows Server 2019.

Step 5: Open the DHCP Console

After installing the DHCP role, I open the DHCP console from:

Server Manager > Tools > DHCP

In the DHCP console, I expand the server and review the IPv4 and IPv6 sections.

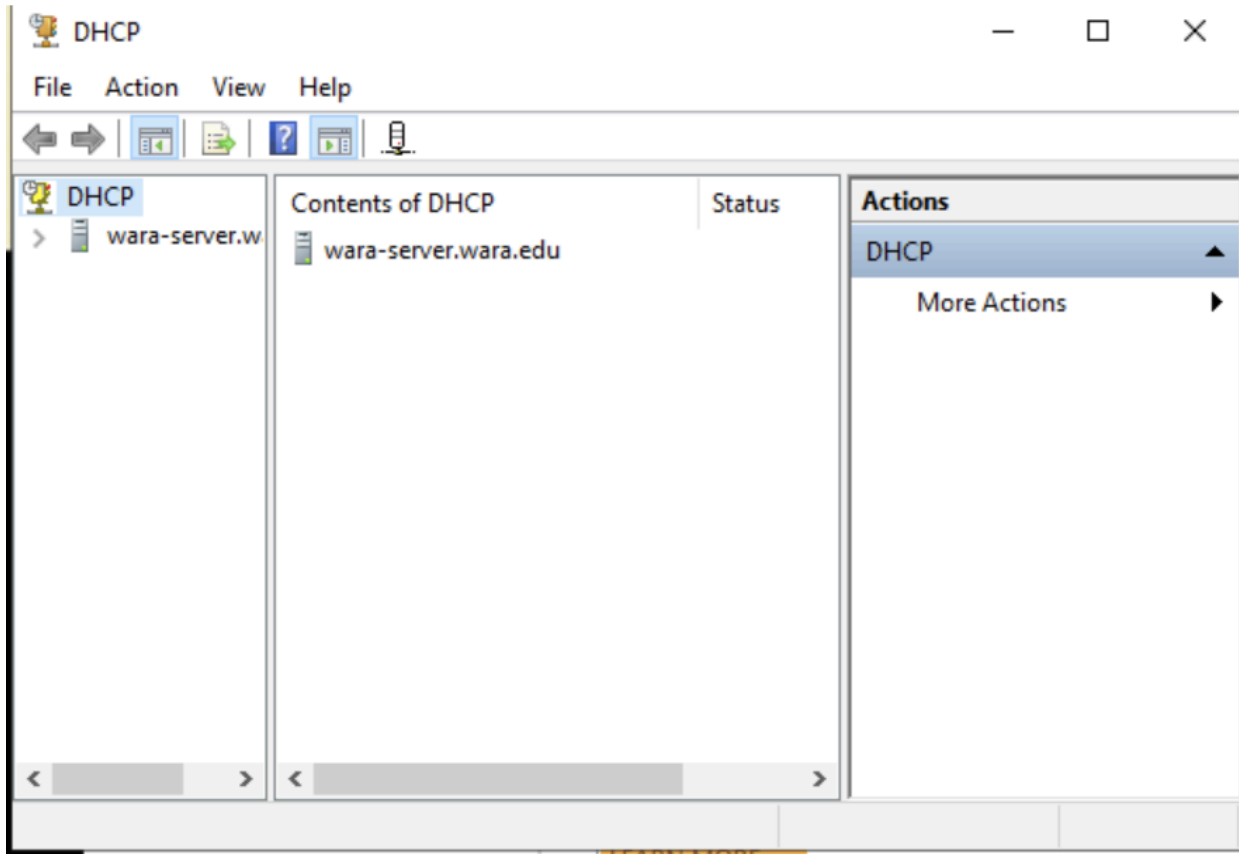


Figure 5: DHCP console opened and server expanded for scope configuration.

Step 6: Create a New DHCP Scope

I create a new DHCP scope under the IPv4 section.

A DHCP scope defines the range of IP addresses that the DHCP server can assign to client computers.

I right-click IPv4, select New Scope, and begin the New Scope Wizard.

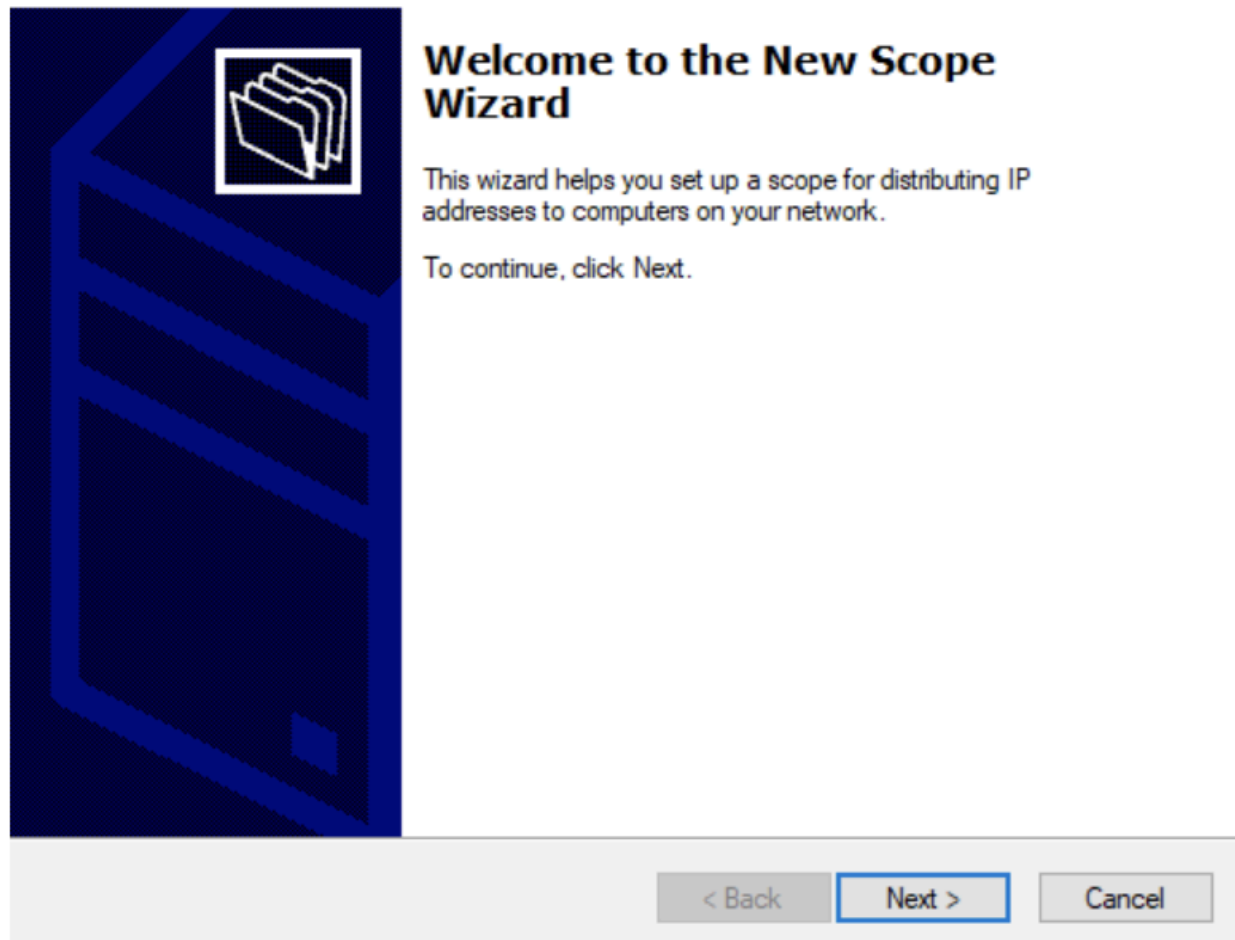


Figure 6: New Scope Wizard started under the IPv4 section of the DHCP console.

Step 7: Configure the DHCP Scope Name

I enter a scope name to identify the DHCP scope.

The scope name helps administrators understand the purpose of the DHCP address range.

New Scope Wizard

Scope Name
You have to provide an identifying scope name. You also have the option of providing a description.

Type a name and description for this scope. This information helps you quickly identify how the scope is to be used on your network.

Name:

Description:

< Back **Next >** Cancel

Figure 7: DHCP scope name configured in the New Scope Wizard.

Step 8: Configure the DHCP IP Address Range

Next, I configure the DHCP scope address range.

The scope uses:

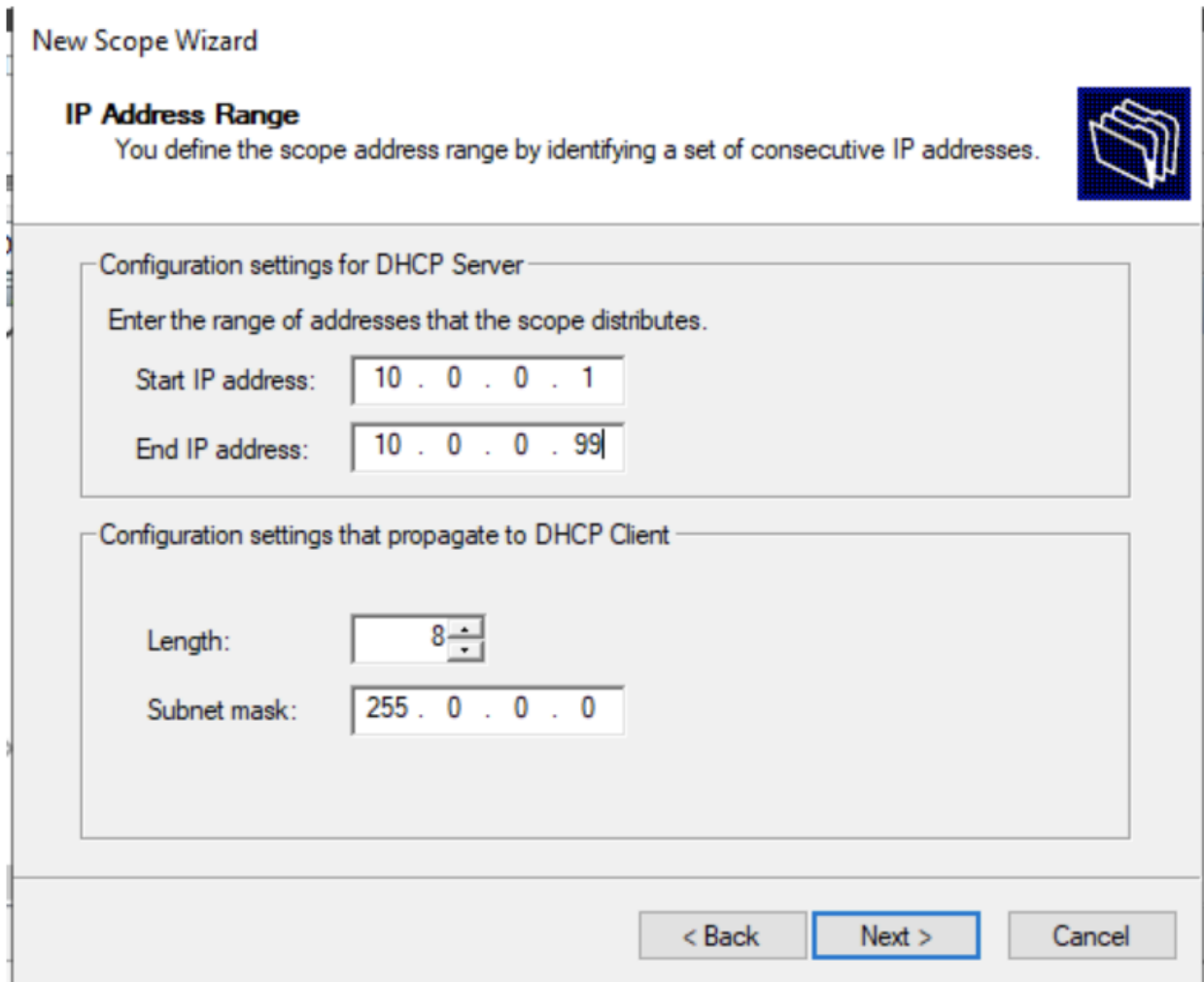
Start IP address: 10.0.0.1

End IP address: 10.0.0.99

Length: 8

Subnet mask: 255.0.0.0

This allows the DHCP server to assign addresses from 10.0.0.1 through 10.0.0.99.



New Scope Wizard

IP Address Range

You define the scope address range by identifying a set of consecutive IP addresses.

Configuration settings for DHCP Server

Enter the range of addresses that the scope distributes.

Start IP address: 10 . 0 . 0 . 1

End IP address: 10 . 0 . 0 . 99

Configuration settings that propagate to DHCP Client

Length: 8

Subnet mask: 255 . 0 . 0 . 0

< Back Next > Cancel

Figure 8: DHCP scope address range configured from 10.0.0.1 to 10.0.0.99.

Step 9: Add an Exclusion for the DHCP Server Address

I add an exclusion for:

10.0.0.52

This prevents the DHCP server from assigning its own static IP address to a client.

The excluded address is important because the server's IP address should not be leased to another device.

New Scope Wizard

Add Exclusions and Delay

Exclusions are addresses or a range of addresses that are not distributed by the server. A delay is the time duration by which the server will delay the transmission of a DHCP OFFER message.

Type the IP address range that you want to exclude. If you want to exclude a single address, type an address in Start IP address only.

Start IP address: End IP address:

Excluded address range:

Subnet delay in milli second:

< Back **Next >** Cancel

New Scope Wizard

Add Exclusions and Delay

Exclusions are addresses or a range of addresses that are not distributed by the server. A delay is the time duration by which the server will delay the transmission of a DHCP OFFER message.

Type the IP address range that you want to exclude. If you want to exclude a single address, type an address in Start IP address only.

Start IP address: End IP address:

Excluded address range:

Address 10.0.0.52

Subnet delay in milli second:

< Back **Next >** Cancel

Figure 9: DHCP exclusion added to prevent 10.0.0.52 from being assigned to clients.

Step 10: Configure Lease Duration

I review the default DHCP lease duration and continue through the wizard.

The lease duration controls how long a client can use an assigned IP address before it must renew the lease.

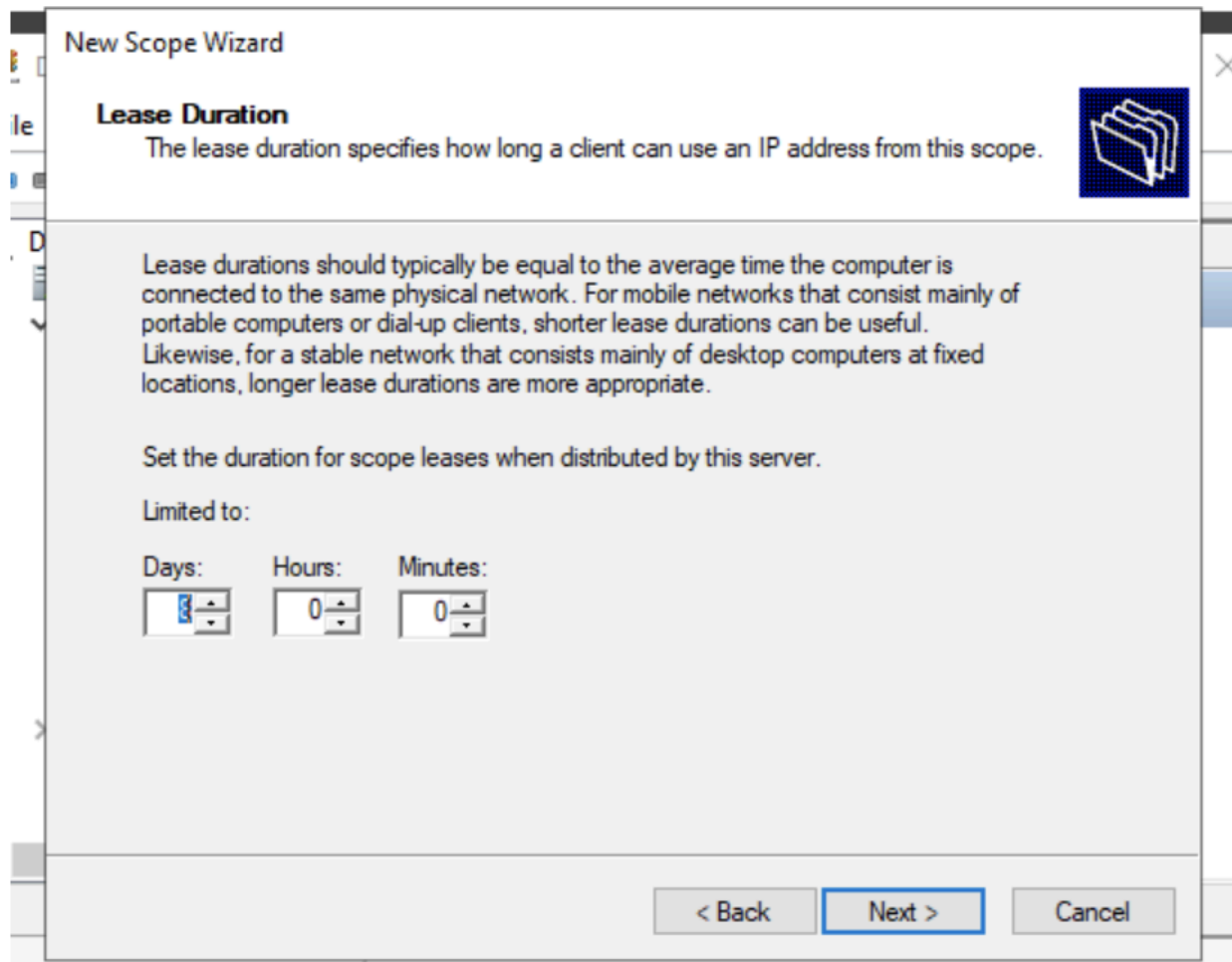


Figure 10: DHCP lease duration reviewed during scope configuration.

Step 11: Configure DHCP Options

I select:

Yes, I want to configure these options now

This allows me to configure common DHCP options such as the default gateway and DNS server.

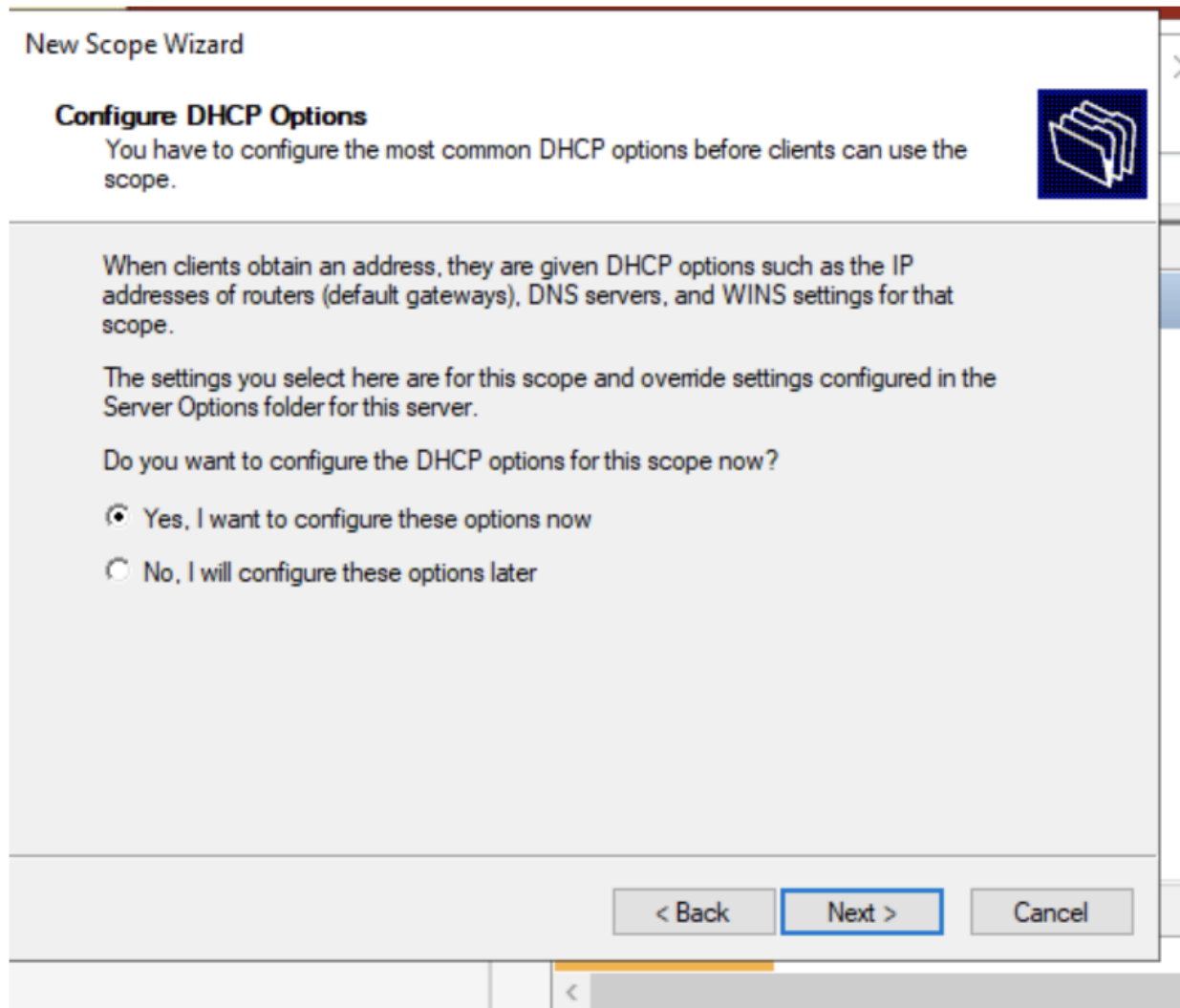


Figure 11: DHCP options selected for configuration during scope setup.

Step 12: Configure Router / Default Gateway Option

I configure the router/default gateway option as:

10.0.0.200

This option tells DHCP clients which default gateway to use for network traffic outside the local network.

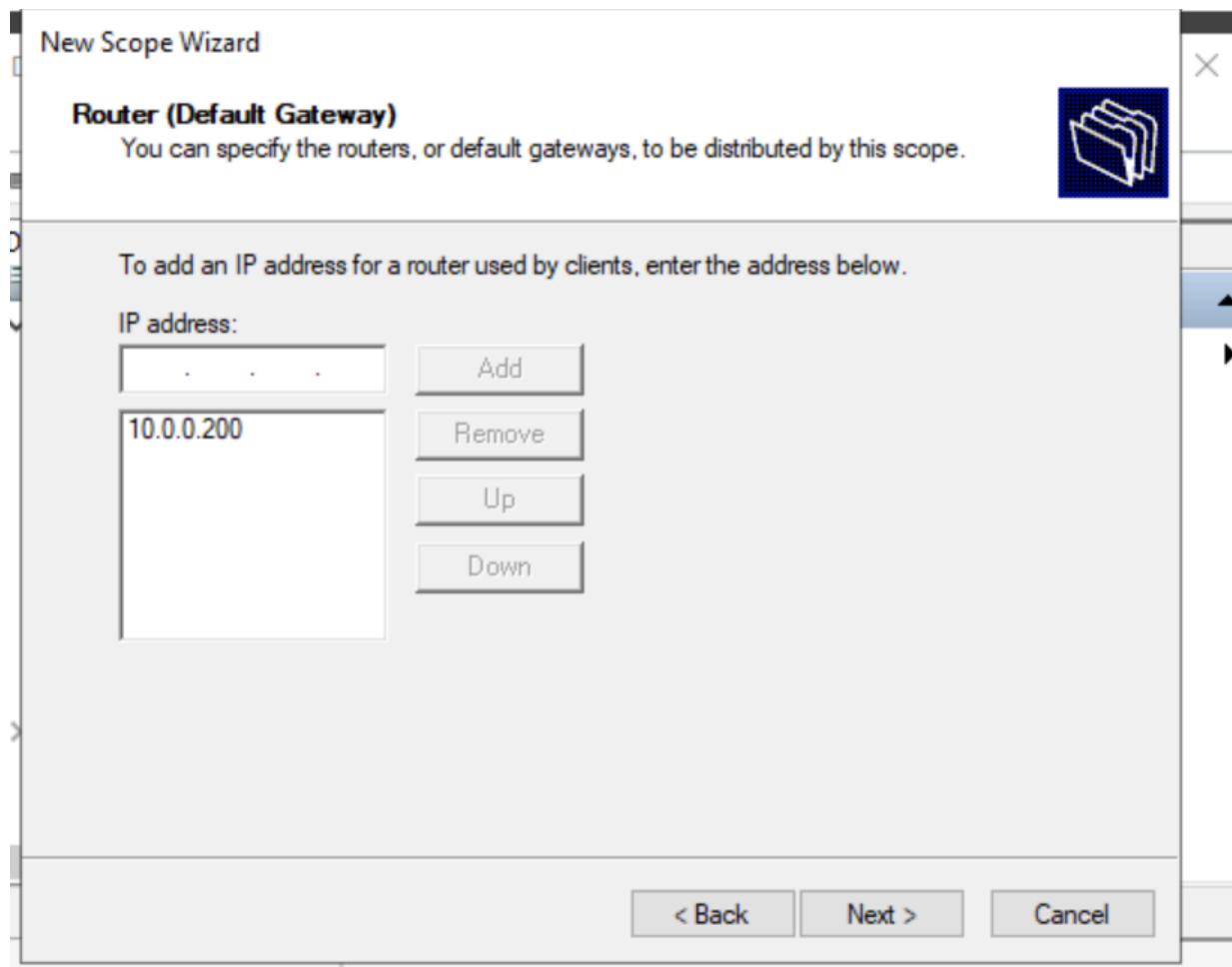


Figure 12: Router/default gateway option configured as 10.0.0.200.

Step 13: Configure DNS Server Option

I configure the DNS server option as:

10.0.0.52

This tells DHCP clients to use the DHCP/DNS server for name resolution.

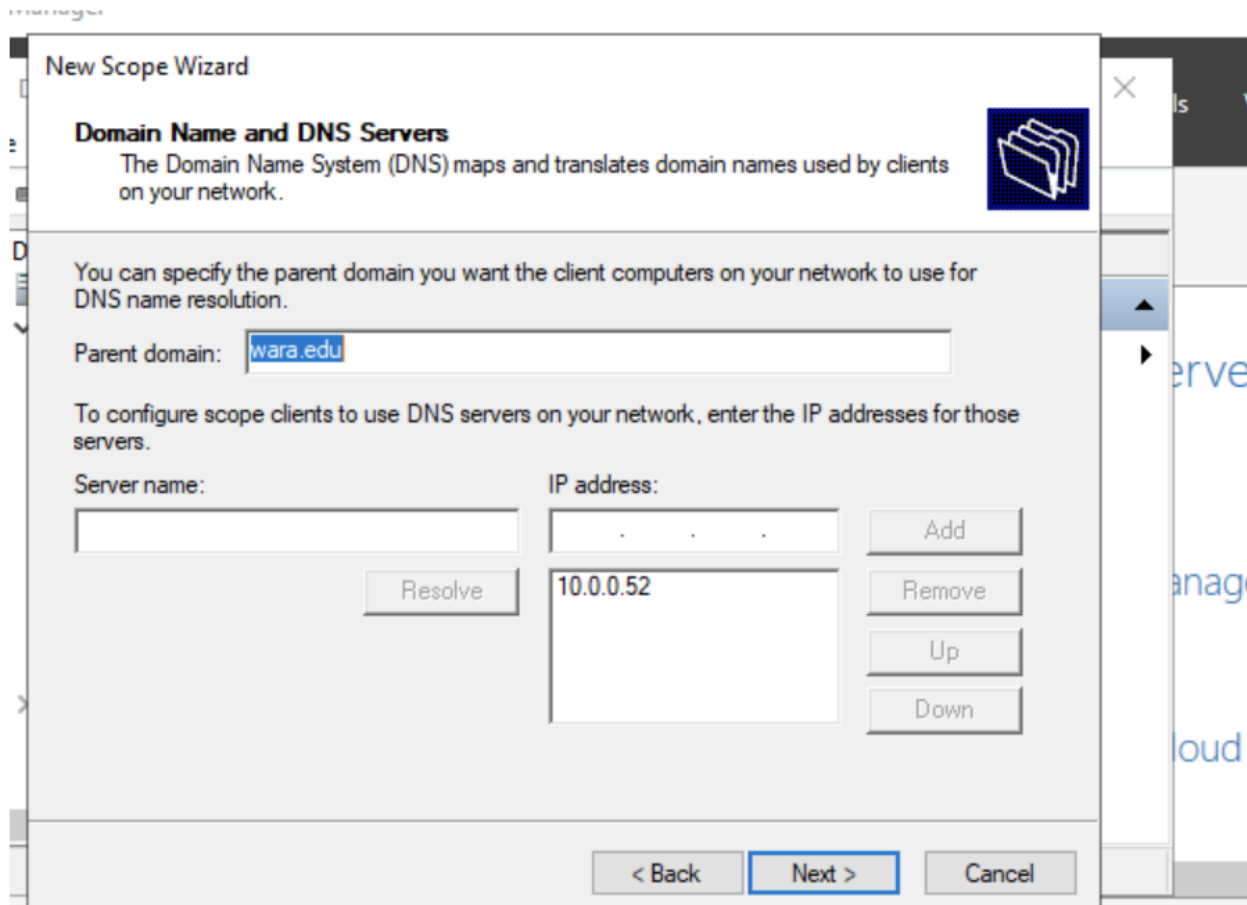


Figure 13: DNS server option configured as 10.0.0.52 for DHCP clients.

Step 14: Activate the DHCP Scope

I activate the DHCP scope so the server can begin assigning IP addresses to clients.

The scope must be active before clients can receive addresses from it.

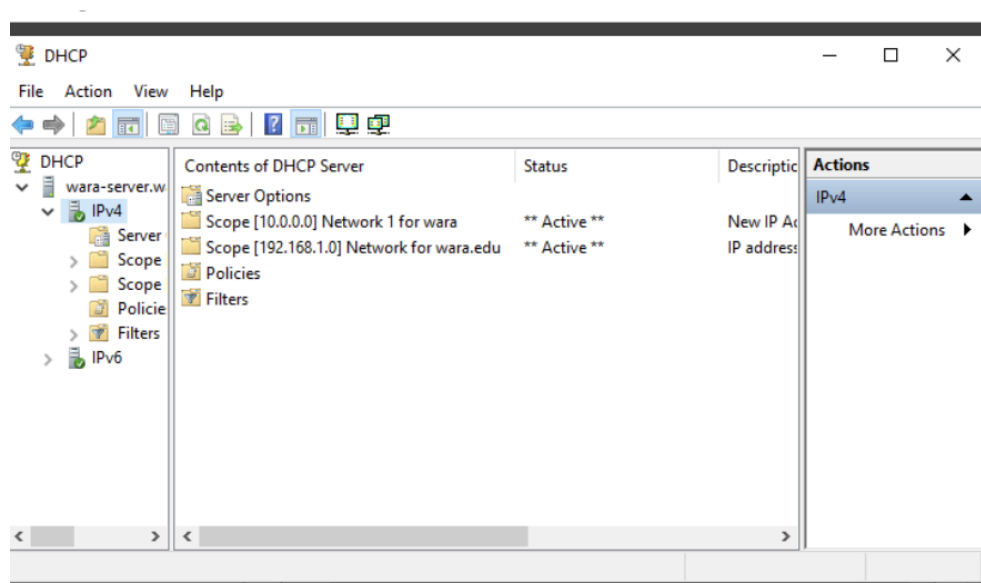
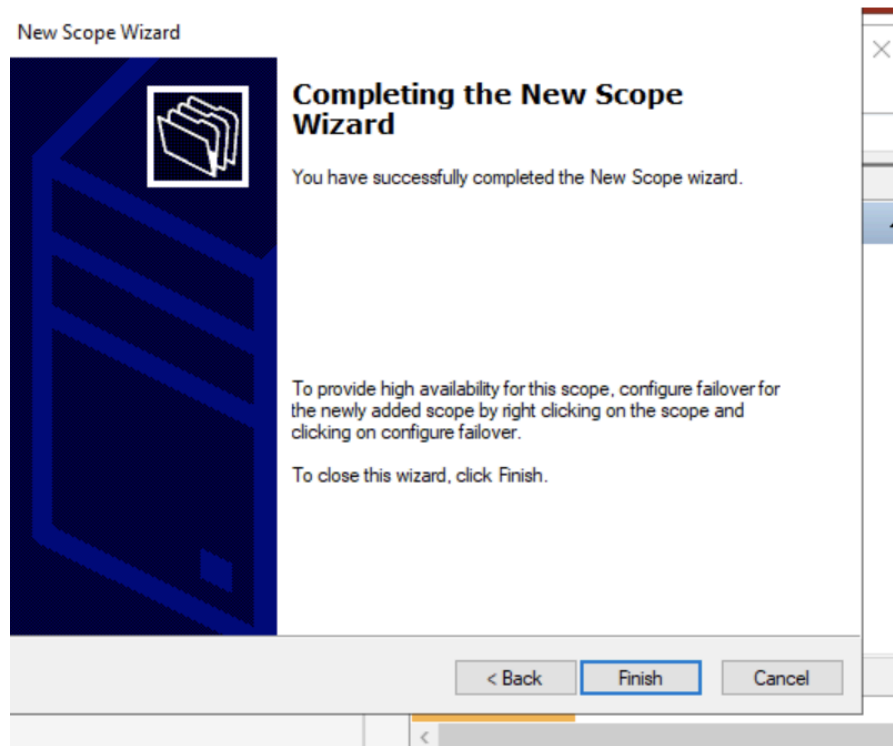


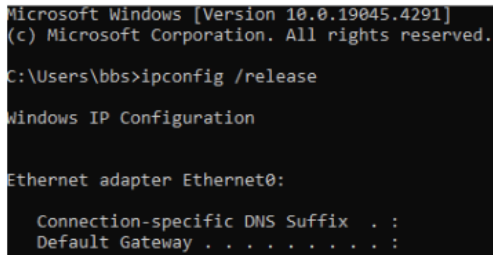
Figure 14: DHCP scope activated and ready to assign IP addresses to clients.

Step 15: Release the Client IP Address

On the Windows 10 client, I open Command Prompt and run:

```
ipconfig /release
```

This releases the current IP address configuration from the client.



```
Microsoft Windows [Version 10.0.19045.4291]
(c) Microsoft Corporation. All rights reserved.

C:\Users\bbs>ipconfig /release

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    Default Gateway . . . . . :
```

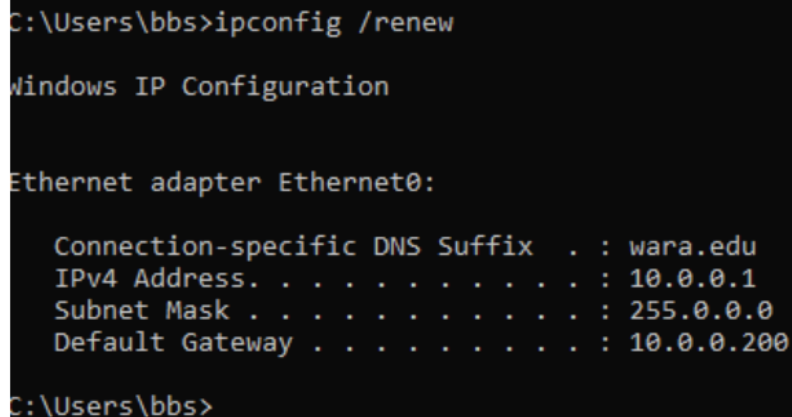
Figure 15: Windows 10 client releases its current IP address using ipconfig /release.

Step 16: Renew the Client IP Address

Next, I run:

```
ipconfig /renew
```

This forces the Windows 10 client to request a new IP address from the DHCP server.



```
C:\Users\bbs>ipconfig /renew

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : wara.edu
    IPv4 Address. . . . . : 10.0.0.1
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 10.0.0.200

C:\Users\bbs>
```

Figure 16: Windows 10 client receives DHCP configuration after running ipconfig /renew.

Step 17: Capture and Analyze DHCP Traffic in Wireshark

I open Wireshark, start a capture, and filter for DHCP traffic.

After running `ipconfig /renew`, I review DHCP messages such as:

DHCP Discover

DHCP Offer

DHCP Request

DHCP ACK

These messages show the DHCP DORA process, where the client discovers a DHCP server, receives an offer, requests the address, and receives acknowledgment.

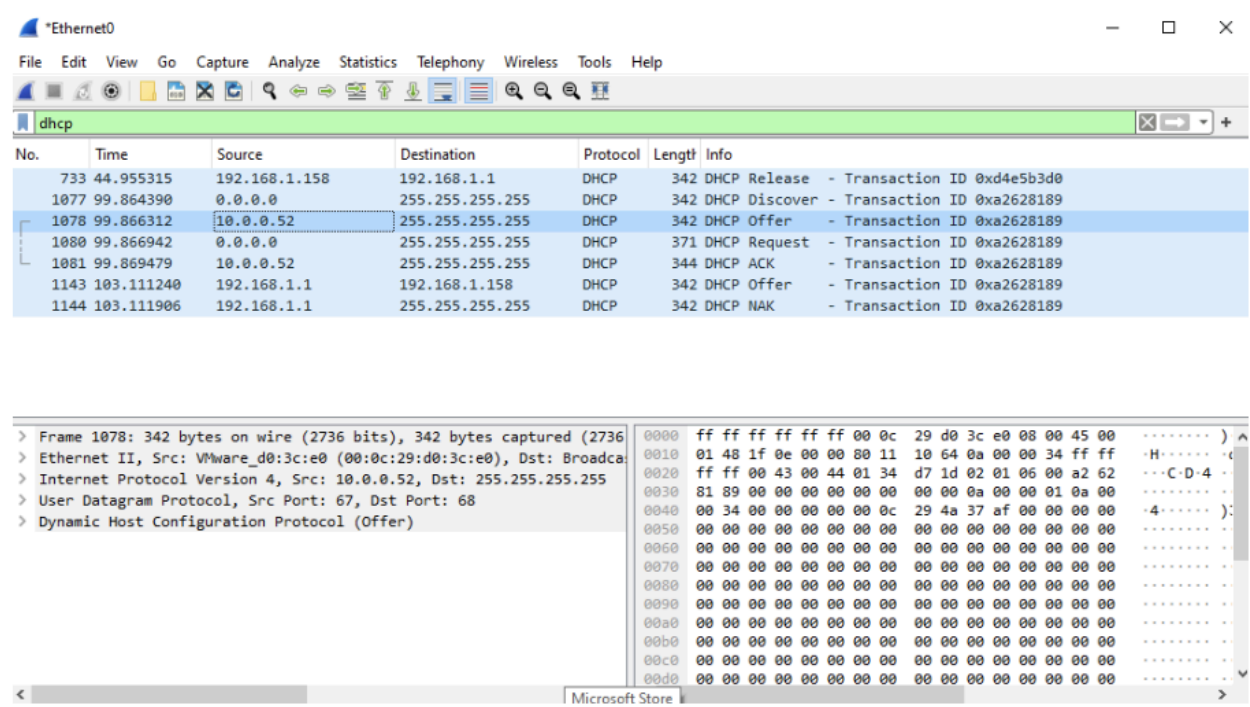


Figure 17: Wireshark captures DHCP Discover, Offer, Request, and ACK traffic during IP renewal.

7. Key Findings

Finding	Explanation
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DHCP clients can receive IP settings automatically	The Windows 10 client is configured to obtain IP and DNS settings automatically.
DHCP servers need static IP addresses	The Windows Server 2019 system uses 10.0.0.52 so clients can consistently reach it.
DHCP scopes define assignable addresses	The scope range is 10.0.0.1 to 10.0.0.99 .
Exclusions prevent address conflicts	10.0.0.52 is excluded so the server address is not assigned to a client.
DHCP options provide network settings	The scope includes gateway and DNS server settings.
Scope activation is required	Clients cannot receive addresses until the scope is activated.
DHCP uses the DORA process	Wireshark shows Discover, Offer, Request, and ACK messages.

8. Analysis

This lab shows how DHCP simplifies network configuration by automatically assigning IP addresses and related network settings to clients. Instead of manually configuring every Windows client, an administrator can configure one DHCP scope and allow clients to receive settings automatically.

The DHCP scope configuration is important because it controls which addresses can be assigned, which addresses should be excluded, and which network options clients receive. The

exclusion for `10.0.0.52` prevents the server's static IP address from being accidentally assigned to a client.

The Wireshark capture helps connect the configuration to actual network traffic. Seeing the DHCP Discover, Offer, Request, and ACK messages shows how the client and server communicate during IP address assignment.

9. IT Support Relevance

This lab is directly relevant to IT Support Analyst, Help Desk, Desktop Support, Network Support, and Junior System Administrator roles.

It connects to real job tasks such as:

- Troubleshooting clients that cannot get an IP address
- Checking DHCP client settings
- Running `ipconfig /release` and `ipconfig /renew`
- Understanding DHCP scopes and exclusions
- Preventing IP address conflicts
- Checking default gateway and DNS options
- Verifying DHCP-assigned configuration
- Using Wireshark to inspect DHCP traffic
- Supporting Windows client/server network environments

10. Skills Demonstrated

This lab demonstrates:

- Windows 10 DHCP client configuration
- Windows Server 2019 DHCP Server role installation
- Static IP configuration
- DHCP console management
- DHCP scope creation
- Scope exclusions
- Lease duration review
- DHCP options configuration
- Router/default gateway option
- DNS server option
- Scope activation
- `ipconfig /release`
- `ipconfig /renew`
- Wireshark DHCP traffic analysis
- DHCP DORA process

- Network troubleshooting documentation

11. Ticket-Style Summary

Issue:

A Windows 10 client needs to receive network configuration automatically from a Windows Server 2019 DHCP server.

Impact:

Without DHCP, clients may not receive valid IP addresses, DNS servers, or gateway information, which can prevent network connectivity.

Steps Taken:

- Configure the Windows 10 client to obtain IP and DNS settings automatically.
- Configure the Windows Server 2019 DHCP server with static IP address 10.0.0.52.
- Install the DHCP Server role.
- Open the DHCP console.
- Create a new IPv4 DHCP scope.
- Configure the scope range from 10.0.0.1 to 10.0.0.99.
- Add an exclusion for 10.0.0.52.
- Review lease duration.
- Configure DHCP options.
- Set the router/default gateway option to 10.0.0.200.
- Set the DNS server option to 10.0.0.52.
- Activate the DHCP scope.
- Run `ipconfig /release` and `ipconfig /renew` on the Windows 10 client.
- Capture and review DHCP traffic in Wireshark.

Root Cause:

The network needs a DHCP scope and DHCP options so clients can automatically receive valid IP configuration.

Resolution:

A DHCP scope is created, configured, activated, and tested from the Windows 10 client.

Verification:

The Windows 10 client receives network configuration after `ipconfig /renew`, and Wireshark shows DHCP Discover, Offer, Request, and ACK traffic.

Status:

Completed successfully.

12. Lessons Learned

In this lab, I learn how DHCP clients and DHCP servers work together to assign IP addresses automatically. The client requests network settings, and the DHCP server provides an available IP address along with additional options such as default gateway and DNS server.

I also learn why exclusions are important. A server's static IP address should not be included in the DHCP pool because that can cause an IP conflict.

This lab helps me understand why DHCP is one of the first services IT support teams check when a user cannot connect to the network or receives an invalid IP address.